Danilee Kelly Visits Greece

Danilee Kelly's quest to see the world's great cow countries took her to Greece, home of Io and Greek yoghurt. She heard that there were many fellow cow enthusiasts in the area, all available on social media. Hence, as her bus pulled into the terminal at Athens, Danilee had two goals on her agenda: to get to her hostel, and to update her MOOSTER account. Preferably, both at once.

Athens is a line of N unit-width columns of varying heights, numbered 1 to N from left to right. The ith column has height h_i. Danilee starts at column 1 and must get to column N.

If Danilee is at column i she is able to move to another column in one of two ways:

- Danilee can *run* from column i to column $i+1$ so long as $h_i \geq h_{i+1}$ (i.e. the next column is not higher than her current one).

  When this happens, Danilee checks in on Mooster and gains f_i followers. She then receives **f** moosages, where **f** is the number of followers she has so far (including the new ones).

- Danilee can *leap* a fixed distance, L, from column i to column $i+L$. She can opt to do so as long as $h_i + H \geq h_{i+1}, h_{i+2}, \cdots, h_{i+L}$ (i.e. the next L columns are no more than H higher than her current one), where H is her (fixed) jump height.

  When Danilee leaps, she gains no followers, however she still receives $L \times \mathbf{f}$ moosages, where **f** is the number of followers she has so far.

  Of course, Danilee cannot leap past column N or else she will land in the Mediterranean.

Your task is to assist Danilee by writing a program to tell her the maximum number of moosages she could possibly receive by the time she reaches column N.

Input

- The first line of input will consist of three space-separated integers, in the format "N H L".
- The following N lines each contain a single integer. The ith of these represents h_i.
- The next $N-1$ lines each contain a single integer. The ith of these represents f_i. Note that the rightmost column does not have an f_i.

Cowtput

Output should consist of a single integer: the maximum number of moosages Danilee can receive by the time she reaches her hostel. If it is impossible for Danilee to reach the hostel by running and leaping as described above, output `-1` instead.

Sample Input

```
6 4 2
4
2
1
4
3
2
1
1
0
1
0
```

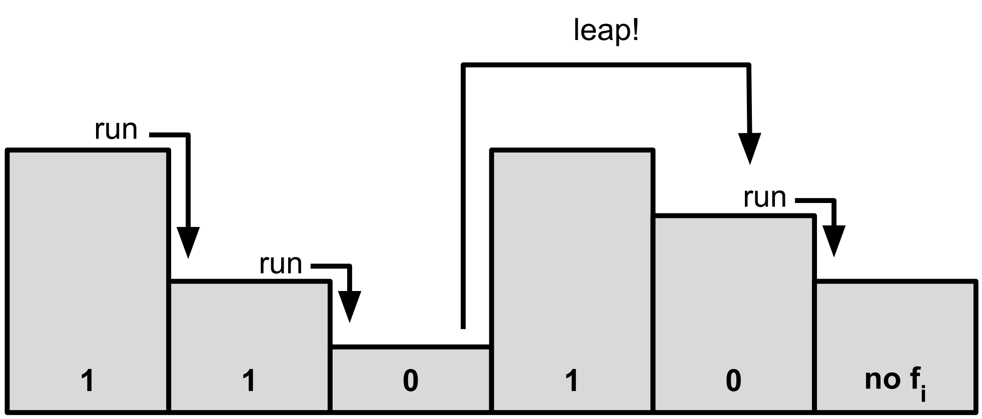
Sample Output

```
9
```

Explanation

- Danilee starts at the leftmost column (height 4) and runs right, gaining 1 follower and subsequently receives 1 moosage.
- She then runs to the 3rd column gaining another follower, so she now has 2 followers and will receive 2 additional moosages.
- At this point she leaps forward $L = 2$ columns to column 5. This gains her no new followers but upon reaching column 5, she still receives $L \times \mathbf{f} = 4$ moosages, where $\mathbf{f} = 2$: the number of followers she has so far.
- Finally, she runs to the hostel at column 6, not gaining any new followers but receiving a further 2 moosages.

This gives a total of $1 + 2 + 4 + 2 = 9$ moosages, which is the maximum possible.



Subtasks & Constraints

For all subtasks:

- $2 \leq N \leq 300\,000$, where N is the width (in columns) of Athens.
- $1 \leq H \leq 1\,000\,000$, where H is Danilee's jump height.
- $0 \leq h_i \leq 1\,000\,000$ for all $1 \leq i \leq N$.
- $1 \leq L \leq 300\,000$, where L is Danilee's jump distance.
- $0 \leq f_i \leq 1\,000\,000$ for all $1 \leq i \leq N - 1$.

Furthermore:

- For Subtask 1 (15 points), $L \leq 10$, $N \leq 1\,000$, and all f_i will be either 0 or 1.
- For Subtask 2 (20 points), $N \leq 5\,000$, and all f_i will be either 0 or 1.
- For Subtask 3 (20 points), all f_i will be 0.
- For Subtask 4 (45 points), no further constraints apply.